M1 Introduction to Crime Scenes Assignment

12 marks

1. One of the first things that investigators do when approaching a crime scene is to make it secure. Give **two** reasons why it's important to properly secure a crime scene. (2 marks)

1) Evidence can be lost or contaminated 2) Anyone can enter the crime scene and disrupt evidence

2. In a crime scene, there are 7 steps that investigators take in order to ensure that protocols are being followed. Complete the table below by describing the 7 steps *in your own words*. (7 marks)

Step	Description
Securing	Make sure no one enters the crime scene; track any who do
Separating	Separate the witnesses so they do not compare stories/speak with each other
Scanning	Decide where photos should be taken by looking over the crime scene
Seeing	Determine where photos should be taken - different angles, different distances of the area and evidence
Sketching	Make a rough sketch of the crime scene; include evidence and landmarks
Searching	Use a spiral, linear, grid, or quadrant search pattern to search the crime scene/area
Securing and Collecting Evidence	Properly package and seal any evidence

3. Describe **two** safety precautions that should be taken at a crime scene other than securing the scene. What would be the possible dangers if these precautions were not taken? (3 marks)

Any 2 of the following:

- Gloves (nitril/latex/heavy) - protection from injury or body fluid evidence; do not contaminate evidence
- Safety helmet - protect head from falling objects
- Shoes - use appropriate shoes to ensure good traction and ankle support; avoid injury
- Eye protection - avoid chemical splashes, fumes or items bouncing into eyes
- Mask - protect from chemicals or odours that can affect the respiratory system
- Protective suit - keeps biohazards off your skin

M2A Individualized and Identified Physical Evidence Assignment

22 Marks

The following scenarios are based upon actual crime cases.

Crime Case 1

One night a hit-and-run collision occurs; a vehicle that lost control at an intersection strikes a parked car. The suspect vehicle speeds away after the collision. Although a license plate number was not obtained, a witness noted that the suspect vehicle was a blue half-ton truck. In addition to fragments of blue paint, police officers find one half of a distinctive plastic license plate frame at the scene and determine that it does not belong to the victim's car. Several hours after the collision, police officers find a damaged blue half-ton truck that has half a plastic license plate frame that matches the fragments of frame found at the earlier collision.

1. Is the license plate evidence individualized or identified? Explain your answer with details from the scenario. (2)

- Identified
- License plate matches truck, but does not specifically tie to driver
- Car could've been stolen

2. What type of physical evidence are the paint fragments? Explain. (2)

- Identified
- Same blue paint may be used in many cars

Crime Case 2

Several red cotton fibres are found upon the ledge of a window that had been broken to gain entry into a home. A person wearing a torn red cotton jacket is apprehended nearby. Tests conducted later show that the fibres from the crime scene and the fibres from the person's jacket have identical physical properties.

1. Explain why the jacket would be considered identified evidence rather than individualized. (2)

- Fiber from the crime scene matches fiber of the suspect's jacket but multiple people could own the same coat
- You could say the tear on the jacket is from the window however it could have been torn before by something else

Crime Case 3

1. State how each of the following groups would have likely categorized (as *individualized* or *identified*) the single hair strand that presented as evidence. Justify your answer for each question. (6)

a. The Prosecution

- The hair presented in this case was considered to be an individualized piece of evidence because the mitochondrial DNA within the hair matched the victim.

b. Scott Petersen's Defense Team

- The hair presented in this case was considered to be identified because the defense thought that mtDNA was not as reliable as nuclear DNA testing.

c. The Jury

- Because Scott Peterson was found guilty beyond a reasonable doubt, the jury must have agreed with the prosecution and found that the hair was an individualized piece of physical evidence.

2. Why was this single piece of physical evidence useful in helping the jury convict Scott Peterson? (2)

- By itself this single piece of evidence may not have been enough to convict Scott Peterson. However, in combination with the other types of evidence presented (i.e., his actions, unusual behavior, taped conversations), it was a powerful piece of physical evidence.

Fingerprint Analysis

1. Can fingerprints be altered or disguised? Explain thoroughly. (2)

- Fingerprints can be slightly altered through damage and scarring, however the individual ridge patterns will still remain the same. The skin on your fingers will regrow with the same pattern as before.

2. How reliable is fingerprinting as a means of Identification? Explain thoroughly and provide specific examples from an actual crime. (2)

- Very reliable, each person's fingerprint is completely unique
- Over 100 different minutae can be identified in fingerprints to match them to suspects
- Give an example from any crime

3. Identify two specific elements of fingerprints that can be analyzed. (2)

- General fingerprint pattern (loops, whorls, arches)
- Other ridge characteristics (i.e. bifurication, islands, etc.)

4. What is a Latent Fingerprint? How are latent fingerprints collected? (2)

- A fingerprint that is not visible to the naked eye but can be enhanced using forensic techniques.
  Methods used:
  - Lifting powder can be applied, the powder adheres to the oils from the fingerprint impression. This can then be removed with tape.
  - Iodine Fumigation. Iodine is placed in a container with a piece of paper containing a latent fingerprint. The iodine combines with the sodium in the sodium chloride (sweat) and creates a dark-blue print.
  - Cyanoacrylate fumigation. This chemical found in superglue can be used to produce a visible white-ish grey film on a latent print. This is especially noticeable on dark surfaces. The resulting print can be photographed. This is useful on solid smooth or lightly textured surfaces.

Fingerprint Analysis Case Studies Assignment

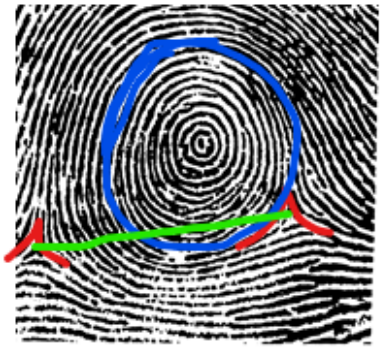
35 marks

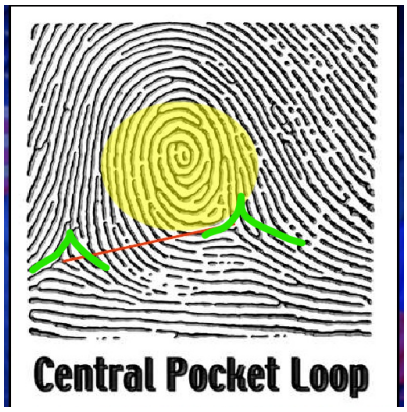
Characteristics of Fingerprints

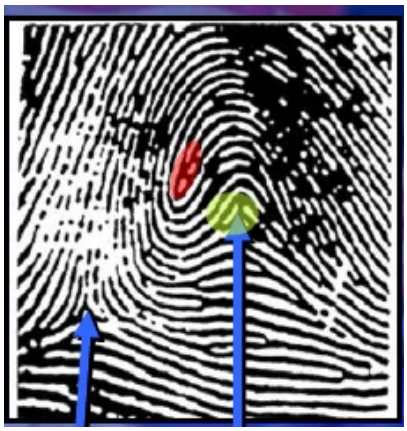
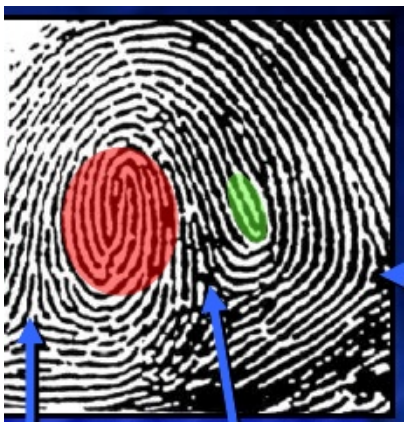
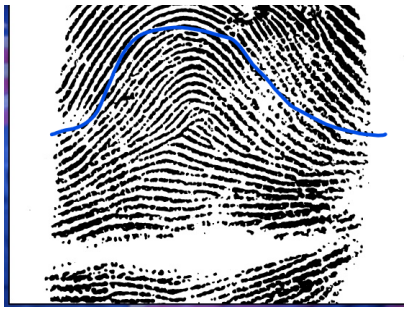
a. Research and define each of the following fingerprint characteristics. Be sure to cite the sources you use (refer back to the course information if you do not remember how to properly reference resources).
b. Find pictures to represent each fingerprint characteristic. On each picture make sure the ridge characteristics are obvious (these will help with identifying fingerprint patterns when looking at a set of fingerprints). See the example provided below.

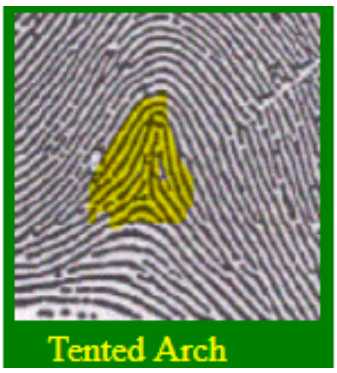
HINT: You can add in the deltas, ridges, and lines using Snip & Sketch or another similar copy and paste app. Once you have drawn on them copy and/or save the picture, then insert it into the table. If you are using a tablet, you can also draw directly on the document. The first one has been done for you:

***1 mark for definition, 1 mark for proper pattern, 1 mark for highlighting key characteristics (deltas, imaginary line, ridge characteristics)

1. Plain Whorl:	A whorl fingerprint pattern that has two deltas and at least one ridge making a complete circuit, which may be spiral, oval, circular, or any variant of a circle. An imaginary line drawn between the two deltas must touch or cross at least one of the recurving ridges within the inner pattern area.	 PLAIN WHORL

2. Central Pocket Loop Whorl (3):	A type of print pattern that has two deltas and at least one friction ridge that makes one complete circuit, which may be spiral, oval, circular, or any variant of a circle; an imaginary line drawn between the two deltas does not touch or cross the "central pocket" (the recurving ridges within the inner pattern area).	
3. Double Loop Whorl (3):	A type of print pattern that consists of two separate loop formations (yellow and green) with two separate and distinct sets of shoulders and two deltas.	
4. Accidental Whorl (3):	A type of print pattern consisting of the combination of two different types of patterns (excluding the plain arch) with two or more deltas; or a print pattern type that possesses some of the	Examples (there are many)

	requirements for two or more different types of patterns; or a print pattern type that conforms to none of the definitions of a pattern.	 
5. Plain Arches (3):	A type of print pattern in which the friction ridges enter on one side of the print and flow out the other side with a rise or wave in the center.	

6. Tented Arches (3):	A type of print pattern similar to the plain arch but that possesses an angle, upthrust (central rise), or two of the three basic characteristics of the loop.	
7. Radial Loops (3): Hand should be indicated R or L	A loop pattern in which the pattern flows in the direction of the radius bone of the forearm, toward the thumb (Remember to flip the hand over to determine this as it changes depending on what hand the print is from.)	Right Hand:
8. Ulnar Loop (3): Hand should be indicated R or L	A loop pattern where the pattern flows in the direction of the ulna bone of the forearm, toward the little finger. (Remember to flip the hand over to determine this as it changes depending on what hand the print is from.)	Right Hand:
Source(s) (2): Use APA or MLA formats Make sure the sources are legit. They should be properly cited, not just a URL.		

APA:
Fingerprint Analysis. (n.d.). Retrieved from http://www.forensicsciencesimplified.org/prints/glossary.html rather than just the URL: http://www.forensicsciencesimplified.org/prints/glossary.html

MLA:
Jury Rocamora Follow. "Whorl Patterns." SlideShare, www.slideshare.net/juroc26/whorl-patterns. rather than just the URL: https://www.slideshare.net/juroc26/whorl-patterns

Now put the information you have found into practice with the following Case Study:

Two suspects wearing ski masks walked into the National Bank in the town of Marioville shortly after it had opened. Suspect #1 passed a note that demanded money to a bank clerk.

Assignment background information omitted - only questions included:

1. Identify the types of fingerprint patterns from the list below that were found on the pieces of evidence given. Assume all prints are from the right hand. (4)

    - Ulnar loop
    - Radial loop
    - Double Loop
    - Tented arch
    - Simple arch
    - Simple/Plain whorl
    - Accidental whorl
    - Central pocket whorl

Evidence	Note given to bank teller	Five dollar bill	Glass #1 from restaurant	Glass #2 from restaurant
Fingerprint Pattern	Ulnar loop	Simple/Plain whorl	Ulnar loop	Simple/Plain whorl

2. Identify the pair of fingerprint patterns as listed below that represents the pair of fingerprint patterns of each of the suspects given below. Assume all prints are from the right hand. (4)

- Ulnar loop
- Radial loop
- Double Loop
- Tented arch
- Simple arch
- Simple/Plain whorl
- Accidental whorl
- Central pocket whorl

Suspect	Joseph Lyons	Doug Banes	Reynold Chapman	Troy Atkins
Fingerprint Pattern 1	Tented arch	Ulnar Loop	Double Loop	Ulnar Loop
Fingerprint Pattern 2	Accidental Whorl	Simple/Plain Whorl	Central Pocket Whorl	Simple/Plain Whorl

3. Why is determining if a set of fingerprints is from the right or left hand important? (2)

- An ulnar loop and radial loop are determined based on the hand
- The print is based on flipping the hand over then looking at where the loop starts and ends - if the loop opens towards the thumb or baby finger

4. Which two suspects do you think are responsible for robbing the National Bank in Morinville? Give **three** pieces of evidence for your decision. (2)

Doug Baines (Suspect 2) and Troy Atkins (Suspect 4)

- Fingerprints match
- Hair color matches
- Height matches suspect description

Trace Evidence Assignment

14 marks

1. Complete the table by giving an example of each of the types of trace evidence that would likely be found at a crime scene. (4)

Type of Evidence	Example
Identified	Footprints, fabric
Individualized	Fingerprints, blood
Organic	Dust, pollen
Inorganic	Gunshot residue

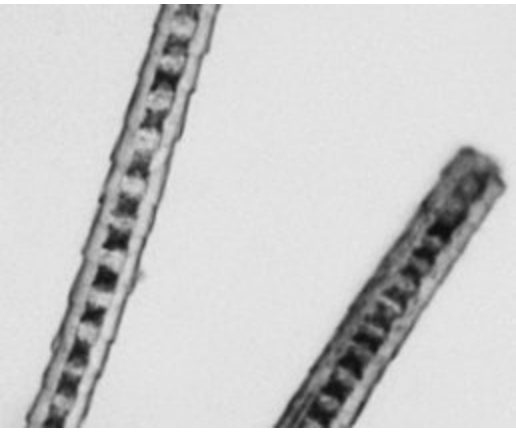
2. Hair and fibre evidence can be valuable in criminal investigations. Is hair and fibre evidence considered identified or individualized evidence? Explain your answer. (2)

- Identified evidence, because hair/fiber analyzed under a microscope cannot be tied directly to a suspect with certainty.
- If the hair's DNA were analyzed, then it would be considered individualized

3. What are some of the limitations of hair and fibre evidence? (2)

- Most human hair looks very similar under a microscope
- Both men and women can have long or short hair
- Some fibers are extremely common

4. The following evidence was found at a crime scene. When it was examined under a microscope, the following image was taken.



a. Give 2 observations about the evidence (2)

- Middle of the strand is visible
- Middle has a stacked pattern (squares stacked in a row)
- Outside of the strand is not smooth

b. Classify the evidence as either hair or fiber. Explain how you came to this conclusion (2)

- Presence of medulla suggests hair and the cuticle is not smooth

c. If you classified the evidence as hair, identify it as either animal or human. If you classified the evidence as fiber, identify it as either natural or man-made. In both cases, explain how you came to this conclusion. (2)

- This is animal hair, because of the stacked medulla

Trace Evidence Case Studies Assignment

Questions:

1. Did the police consider the hair evidence collected from the teenagers to be identified or individualized evidence? Explain. (2)

- The police considered the evidence to be individualized. They believed that the hair evidence could be directly linked with the suspects.

2. Why is it important that microscopic hair comparisons be confirmed by DNA determinations? (2)

- Many human hair samples are too similar in appearance to be individualized. Hence, DNA should be used when attempting to individualize hair evidence.

Questions:

1. What led investigators to conclude that the bandage found at the crime scene belonged to the suspect and not to the victim and that the suspect belonged to the military? (2)

- No blood from the victim was found, so the bandage did not belong to the victim
- A military disinfectant was found on the bandage

2. Describe the fibre evidence that was found to support the case against Samuel Morgan. (2)

- Morgan's sister-in-law's bandage fiber matched the fiber from the bandage found at the crime scene

Body Fluid Evidence Assignment

21 marks

1. There are many different types of body fluid evidence that would be useful at a crime scene. Complete the table below to outline 4 types of body fluid evidence. (8 marks)

Type of Evidence	Why it is useful	Type of crime where this evidence would appear
Blood	Determining the blood type of the victim or suspect	Violent crime, murder, assault
Saliva	If the victim is a secretor, this can be used to determine their blood type	Sexual assault
Semen	This can be evidence of sexual assault. DNA can be taken from semen	Sexual assault
Skin	Can be linked with injuries on the victim or suspect. DNA can be taken.	Abduction, unlawful restraint

2. There are many tools that forensic scientists use to analyze body fluid evidence at a crime scene. Complete the table below to demonstrate your knowledge of these techniques. (8 marks)

Tool	How it is used	Why it is used	Advantages	Disadvantages
Phenolphthalein	Turns pink if in contact with hemoglobin	To determine if an unknown substance is blood	Fast Very sensitive	False positives
Luminol	Glows when sprayed on blood	To find trace amounts of blood that has been cleaned	Can be used after a long time Does not affect other tests Very sensitive	False positives

3. Police suspect that there was a violent murder committed inside a house and that the culprit cleaned up after the attack before burying the body. Suggest possible areas within the house that investigators could check for small traces of blood that the culprit may have missed while cleaning up. (3 marks)

- Underneath carpets or floorboards
- Splatter on the ceiling
- Under baseboards
- Inside drains or sinks

4. Explain how being able to prove that a suspect is a 'secretor' be helpful to investigators? (2 marks)

- Secretors are people who show their blood type in other body fluids besides blood. This information can be used to tie a suspect to a crime, especially if the suspect has a rare blood type.

Body Fluid Case Studies Assignment

28 marks

Crime Case Study #1: Unconvincing Circumstantial Evidence

1. Explain how the blood type of the killer could be determined from the cigarette butts found at the crime scene? (2)

The killer was a secretor, meaning their saliva can determine their blood type

2. Why was the determination that the killer of Walter Dinivan had blood type AB+ an important piece of evidence? (2)

The AB+ blood type is very rare - only found in ~3% of the population

3. Identify the physical evidence found at Walter Dinivan's home as either Identified or Individualized crime scene evidence. (4)

Evidence	Identified or Individualized
Paid off bank loan	Individualized
Brown paper bag	Identified
Cigarette butts	Individualized
Fingerprint on glass	Individualized

Crime Case Study #2: Which Neighbour is a Killer?

1. Why does the blood spatter residue indicate that it was unlikely that Graham Backhouse had been pursued by Colyn Bedale-Taylor in the hallway? (2)

The blood stain pattern indicates that the blood fell from a stationary place. Colyn was likely not moving quickly.

2. What type of blood spatter pattern should have appeared in the hallway if Graham Backhouse had shot Colyn Bedale-Taylor in self-defense as he claimed? (2)

The blood would be in smaller droplets and be more spread out. The blood would also likely form teardrop shaped droplets.

3. Explain what the blood spatter evidence in the hallway of Graham Blackhouse's home indicated about the events. (2)

Graham Blackhouse likely stabbed himself and then shot Corlyn.

Crime Case Study #3: The Death of Tammy's Parrot

1. Explain why police became suspicious of Tammy's story after discovering she had only a few superficial cuts on her lower arms. (2)

Very large amount of blood that could not have come from Tammy's cuts; neighbours indicated they had not seen Tammy's ex-husband for over a month and did not overhear any arguing on the day in question

2. Describe the most probable source of the majority of blood smeared upon Tammy and on her apartment walls. (2)

From her parrot that had been stabbed to death and hidden under the bed

3. Explain how microscopic analysis of the blood smeared at the crime scene could help to prove the source. (2)

Under a microscope, red blood cells from humans are anucleated (no nucleus) but red blood cells from a bird would have a nucleus

Crime Case Study #4: How did Phil Beardman die?

1. What does the blood spatter evidence found on the wall above the stairs in Phil's condo indicate? (2)

Phil was likely hit by a blunt object before falling down the stairs - the droplets on the floor and edge show a medium-velocity impact blood spatter - the strands of blood indicate that he was moving after being hit

2. From the list of possible events below, order the most likely sequence of events that occurred on the night Phil died from 1 (first) to 4 (last). You do not need to use all of the events. (4)

Possible Event	Order (1 to 4)
Susan pushed Phil down the stairs.	4
Phil began bleeding from his injuries.	3
Phil and Susan had a loud argument.	1
Phil accidentally slipped and fell down the stairs.	
Susan hit Phil with an object at the top of the stairs.	2
Susan hit Phil with an object at the bottom of the stairs.	

3. Explain why the blood spatter evidence indicates that it is unlikely that Phil had fallen down the stairs by accident as Susan suggested. (2)

There were blood spatter patterns that indicated free-falling blood which would mean that Phil would have had to have been standing at the top of the stairs after he was hit. The strands of blood indicate that while he was bleeding he was pushed down the stairs - indicates movement towards where he landed at the bottom of the stairs where the blood started to pool.

Impaired Driving Assignment

27 marks

1. Complete the following chart that summarizes the major physiological and behavioral changes in the human body when exposed to increasing amount of alcohol. (10)

BAC (mg/100mL)	Symptoms	Part(s) of Brain Affected
Euphoria (30-120)	• more self-confident • slow reaction time • poor judgment • shortening of attention span • flushed face • difficulty with fine movements	Cerebrum
Excitement (90-250)	• drowsiness • difficulties understanding or remembering • increasingly slower reaction time • poor coordination and loss of balance • blurred vision	Cerebrum, Cerebellum
Confusion (180-300)	• confusion; dizziness and perhaps staggering; cannot see clearly • highly emotional--aggressive, withdrawn, or overly affectionate	Cerebrum, Cerebellum, Hypothalamus

	• slurred speech; uncoordinated movements; high pain tolerance	
Stupor (250-400)	• hardly move at all; cannot respond to stimuli; cannot stand or walk • vomiting; lapses in and out of consciousness	Cerebrum, Cerebellum, Hypothalamus, Medulla Oblongata
Death (350-500)	• unconscious; reflexes are depressed (for example, pupils do not respond). • breathing slow /shallow; lower than normal body temperature; heart rate may slow.	All parts are affected

2. Identify two specific symptoms caused by alcohol consumption that impair a person's ability to drive. Explain, in detail, how these two symptoms impair a person's ability to drive safely. (4)

- Reaction time is slower, person is unable to respond quickly to changing road situations
- Poor judgement, person is unable to make correct driving decisions
- Difficulty with motor movements, lack of coordination makes mechanics of driving difficult
- Drowsiness, person's ability to react is slower
- Blurred vision, unable to see hazards/pedestrians

3. State the BAC that defines a person as intoxicated under the law and guilty of impaired driving in Canada. (1)

0.08 mg%

4. Given the information about the symptoms of alcohol consumption at various BACs, do you feel that the legal limit for impaired driving is too low, too high or just right? Justify your answer. (2)

Answers may vary

Polygraphing and Forensic Writing Analysis Assignment

15 marks

1. Explain **three** ways that a polygraph examination can help in the police investigation of a crime. (3 marks)

- Detect when a suspect is not being truthful
- Cause a suspect to be anxious enough to confess
- If a suspect refuses to take one, it can indicate potential guilt

2. a) Identify the type of person who may lie repeatedly but who will very likely pass a polygraph examination. (1 mark)

- A compulsive liar

b) Explain how this type of person is able to pass polygraph examinations. (1 mark)

- They are so accustomed to lying compulsively that they do not exhibit the same anxious symptoms that the polygraph detects
- They can fool themselves into believing their own lies

3. The Supreme Court of Canada in 1987 ruled that polygraph evidence cannot be used in a criminal trial.

a) Using your knowledge of polygraphing, give two reasons why polygraph evidence is not admissible in court. (2 marks)

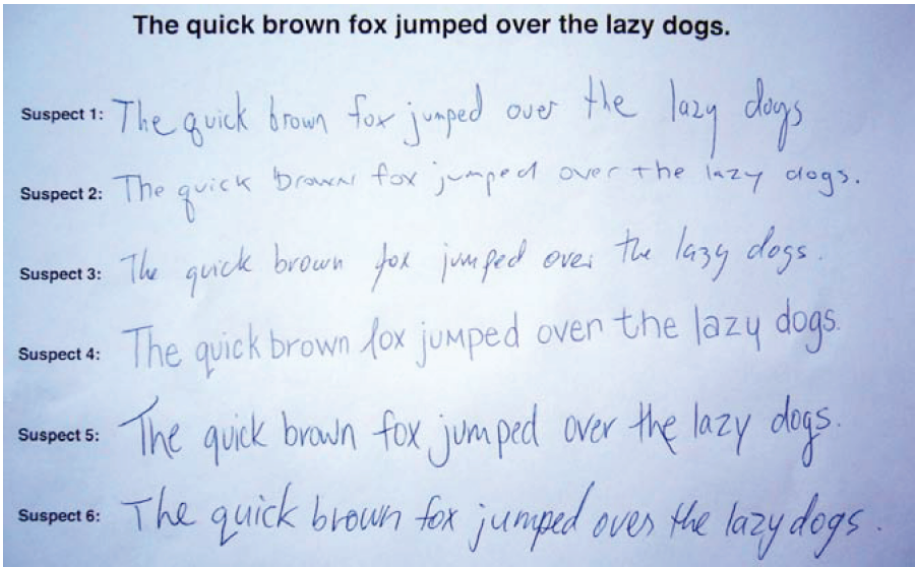
- The results can be misinterpreted, they are often subjective and based on the experience and skill of the examiner
- The polygraph cannot detect all lies
- The polygraph may detect when someone is lying because they are simply anxious about certain topics

b) Do you agree with the Supreme Court's decision to disallow polygraph evidence? Explain why or why not. (2 marks)

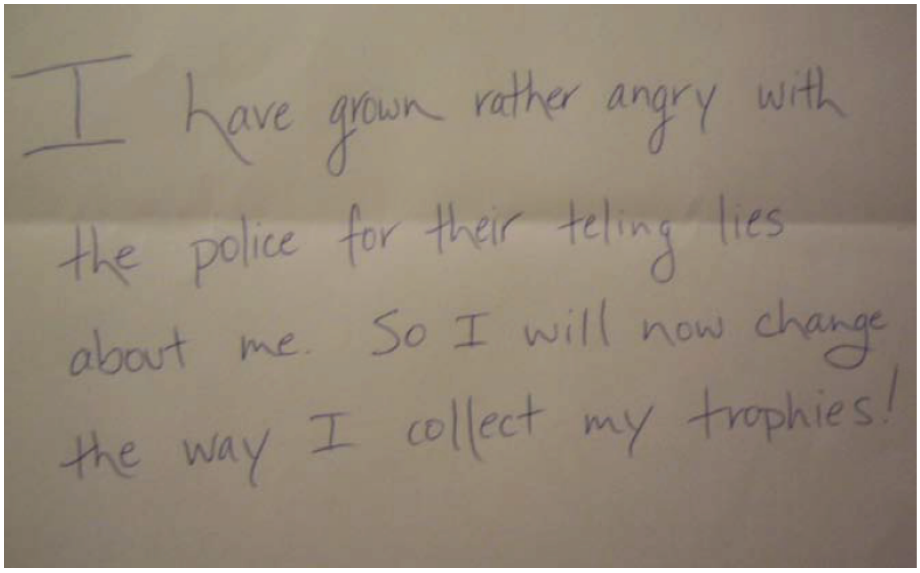
Answers may vary

4. Police are attempting to identify the writer of a threatening note. Below are the writing samples from 6 possible suspects.

Photograph A: Writing Samples from Six Subjects



Photograph B: Threatening Note



Which suspect do you think is most likely to have written the note? Give two pieces of evidence that indicate this. (3 marks)

- Suspect 5
- Long tail on the end of 'h'
- Full loop on 'g'
- Shape of letter 'y' (two lines)
- Vertical slope of letters

5. Explain three specific ways that forensic experts try to minimize the alteration of a handwriting sample by a criminal suspect. (3 marks)

- having the suspect write several pages
- using the same or similar type of paper as the suspect document
- using the same or similar writing instrument as used in the suspect document
- having the suspect use many of the words and sentences found in the suspect document
- not providing any assistance with spelling, punctuation, or with use of lower or upper case letters
- telling the suspect what to write (verbal dictation)
- having the suspect write the same verbal dictation three separate times

Polygraphing and Forensic Writing Case Studies Assignment

14 marks

Crime Case Study #1: Nine Long Days - The Susan Smith Case

1. Give two pieces of case evidence that were determined from polygraph testing that helped the police in solving this crime. (2 marks)

- David's results indicated that he did not know about the disappearance
- Susan's results indicated that she was being deceptive

2. Using your knowledge of polygraph tests, give two reasons why Susan Smith's polygraph results were inconclusive even though she was guilty. (2 marks)

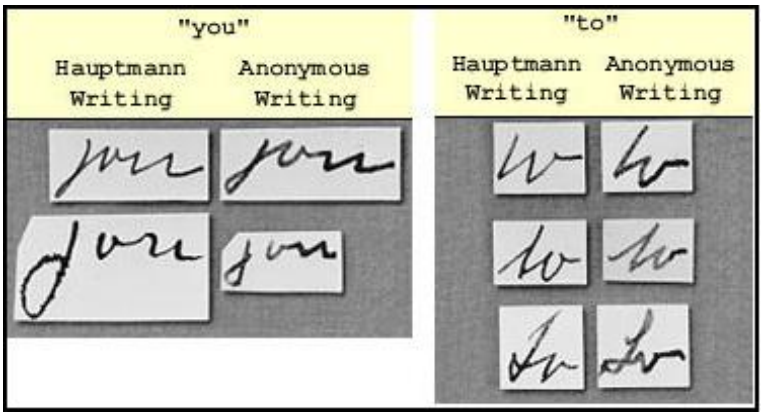
- She may have been nervous throughout the test, meaning her anxiety levels did not differ sufficiently when lying or being truthful
- She may be a compulsive liar
- She may have truly believed her lies
- The officer conducting the test may have been inexperienced

Crime Case Study #2: Kidnapping of the Lindbergh Baby

1. Using your knowledge of graphology, give **two** possible reasons that led the police to believe that the ransom notes were written by the same person. (2 marks)

- The letters may have had a consistent slant
- The letters may have been written with the same type of pen
- The letters may have had unique features that remained consistent

2. In the writing samples below, describe the distinct features of **two** letters that indicate the samples were written by the same person (2 marks)



- The letter y
- The letter t is not crossed
- The letter o is open (like a u)

Forensic DNA Evidence Assignment

18 marks

1. Complete the chart below to analyze the benefits and drawbacks of both RFLP and PCT methods of DNA print analysis. (4)

	RFLP	PCT
Cells needed	5000 to 50 000 cells	50 cells
Time required	Time consuming	Much faster

2. Explain how VNTR and STR sections of DNA used in forensic DNA analysis. How are investigators able to determine a DNA match using these techniques? (2)

- Investigators examine “junk” DNA, which are the VNTR or STR segments.
- By examining DNA at the same locus, investigators can determine if the segments are a match
- By matching 5 different loci, they can determine conclusively that the DNA samples are matching. The chances of two different people having the same VNTR or STR at 5 different places is extremely unlikely.

3. Explain the main differences between nuclear DNA and mtDNA. Give an example of when each would be used. (3)

- Nuclear DNA is inherited from both parents whereas your mtDNA is inherited from your mother and will be identical to hers (all children who share the same biological mother have the same mtDNA).
- Nuclear DNA is contained in the nucleus of the cell, where as mtDNA is contained in the mitochondria of the cell
- DNA will be used when it can be extracted from an intact cell nucleus, mtDNA can be extracted from ancient or badly damaged remains

4. Why is it important to include a probability ratio with every DNA print match? (2)

- This determines the strength of the match (how likely the DNA is to be an actual match). Example: a probability ratio of 1:55 000 000 would mean one other person in 55 million could have the same DNA pattern - this would be a

strong match. The more chance that others could be matched means the match isn't as strong (weak ratio 1:1200 vs strong ratio 1:4 000 000 000.

5. a) Complete the chart below to outline the benefits and drawbacks implications of having a comprehensive national or global DNA data bank. (2)

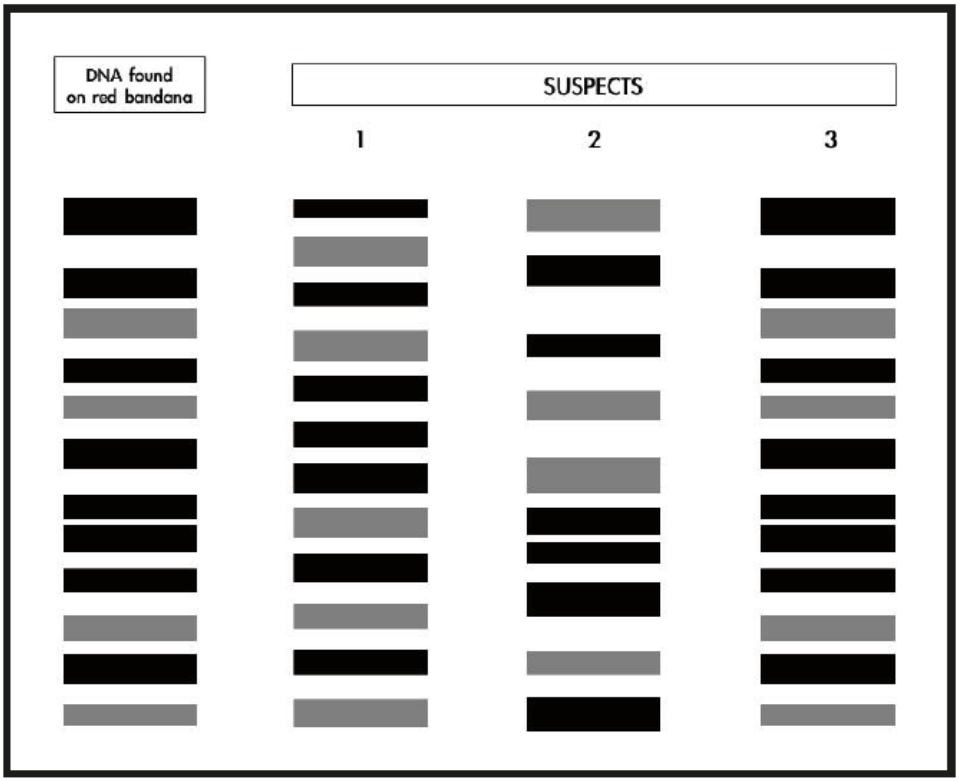
Benefits	Drawbacks
• If DNA is found at a crime scene, it can be quickly matched with a suspect in the database	• Information could be hacked • Time consuming to collect all the data

b) Are you in favor or against having a comprehensive national or global DNA data bank? Give at least 2 reasons for your answer. (2)

Answers will vary.

Fictional Case Study: The Bandana Bank Robbery

1. The DNA from the red bandana as well as from each of the suspects was analyzed. The results are below. From this evidence, identify the suspect whose DNA matches the profile created from the hair found in the red bandana. (1)



Suspect 3

2. Due to the fire, only 60-75 intact cells were found in each of the bandanas. Which type of DNA analysis method will likely be required to analyze this limited number of cells - PCR or RCLP? Justify your choice. (2)

PCR; RFLP requires numerous cells to run an analysis (5000 to 50000 cells) where as PCR only requires about 50 cells to run an analysis; PCR also takes a fraction of the time to process the DNA samples vs. using RFLP